What Reprices and How Fast in Weighted-Average-Maturity Pass-Through

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Abstract

Weighted average maturity is a natural first statistic for judging how quickly a rate shock reaches a sovereign interest bill. It is also easy to overuse. Maturity describes when debt leaves the stock; not all repricing waits for that event. This paper separates the Portuguese debt stock into maturity-driven, reset-driven, and retail-flow components and asks how wide the plausible pass-through range becomes once those clocks are separated. The paper does not treat the behavioural coefficient as a central correction. The competing-return spread does carry a detectable positive association with retail repricing, but the falsification test loads at $p = 0.07$ on a statistic no household observes, so the association is reported as descriptive rather than identified; the bootstrap interval for asymmetry is [-0.016, +0.046]; and conditional historical validation gives no stable forecasting claim. In the current generated tables the scenario kernel is ahead at 2 of three cut dates, but the margins are small and have moved with implementation corrections. The raw episode is nevertheless informative, because households moved money into certificates during the 2022–2023 tightening rather than redeeming them. The defensible result is a bounded sensitivity exercise. Portuguese aggregate data are enough to show that weighted average maturity does not pin down one-year pass-through, but not enough to identify a single behavioural correction or a single fiscal euro amount.

1 Introduction

When interest rates rise, the public debate usually moves quickly from the market yield to the budget line. That leap hides the part of the problem that matters most for annual fiscal accounts: the old debt stock does not wake up one morning carrying the new rate. Some of it matures and is refinanced; some of it resets while staying in place; some of it is retail money that may flow in or out when the offered return changes. Weighted average maturity is a useful summary of the first channel, but it is often asked to do the work of all three.

The shortcut is attractive because it is clean. A portfolio with average maturity m can be treated as if each remaining euro has a constant probability $1/m$ of repricing every year. The companion paper uses exactly that device in its refinancing section because it makes the timing assumption visible. This paper keeps the same question and removes the homogeneity assumption. The object of interest is not the average date at which a bond matures; it is the share of the opening stock that has actually received a new rate by each horizon.

That distinction changes the interpretation of a rate shock before any behavioural story enters. Floating-rate and inflation-linked liabilities do not need to leave the stock in order to reprice. Retail instruments can also add new money at the prevailing rate, which is economically similar to repricing from the budget's point of view even when it is not redemption and replacement. A maturity-calibrated hazard cannot see either channel. It treats repricing and retirement as the same event, which is why it has trouble with portfolios that contain meaningful reset and retail blocks.

The Portuguese case is useful because those blocks are observable enough to separate. IGCP publishes monthly instrument-class stocks and portfolio indicators, and the 2022–2023 tightening created a natural stress test for retail savings certificates. The raw pattern is already informative: households moved money into certificates when short rates rose, because the certificate remuneration formula became attractive relative to bank deposits. The behavioural channel therefore runs through subscriptions, not through the redemption margin that a simple "holders flee to higher rates" story would suggest.

The main result is a measurement warning rather than a forecast. Weighted average maturity is not enough to determine one-year pass-through once reset instruments, retail instruments, and maturity-driven debt are separated. The central behavioural effect is set to zero in the kernel tables. That is not because the spread coefficient is small — it is statistically detectable — but because a falsification test on a statistic no household observes also loads at $p = 0.07$, which points to common time variation the specification cannot separate from the spread. The bootstrap interval for asymmetry covers zero, and the backtest establishes no general forecasting gain over the maturity benchmark. Behavioural responses are therefore carried as sensitivity bounds rather than as a central correction.

The paper's practical message is correspondingly modest. Weighted average maturity should not be discarded; it should be used for the part of the stock whose repricing is genuinely tied to maturity. Reset instruments and retail flows need their own clocks. Once those clocks are separated, the fiscal burden also needs an explicit GDP path, because the same euro interest bill can look quite different as a share of GDP.

The contribution is not the conceptual distinction between maturity and refixing. Debt managers already make that distinction. The contribution is the Portugal-specific measurement exercise: a WAM-calibrated pass-through proxy is compared with the debt manager's published refixing-risk windows at the same date, with a composition-aware scenario, and with a fiscal burden translation that separates physical refixing from shock-weighted pass-through.

2 Related literature

The paper sits between debt-maturity theory, sovereign pass-through accounting, and household-deposit pass-through. The maturity literature explains why the term structure of public liabilities matters for fiscal risk. Missale and Blanchard [11] make debt maturity central to the burden of government debt. Angeletos [3] shows how a non-contingent maturity structure can replicate state-contingent fiscal payoffs in a complete set of maturities. Greenwood, Hanson and Stein [8] study maturity choice as a comparative-advantage problem for government debt management. Arellano and Ramanarayanan [4] and Broner, Lorenzoni and Schmukler [5] show, in different settings, that maturity is not a neutral technical detail: it interacts with spreads, crisis conditions, and rollover risk.

This paper does not estimate an optimal maturity structure and does not model default. Its narrower concern is the measurement step that often comes before those questions. If a sensitivity exercise says that a rate shock reaches the budget according to weighted average maturity, it has already assumed that the portfolio is homogeneous enough for one maturity clock to stand in for all repricing clocks. The Portuguese data give a concrete way to check that assumption.

The distinction itself is standard in debt-management risk practice. Jonasson and Papaioannou [10] describe Average Time to Re-fixing as a sovereign debt-portfolio interest-rate-risk measure: fixed-rate debt is exposed when it is refinanced, while floating-rate debt is exposed when coupons reset. The ESDM formula document [7] formalises the same logic for Average Time to Next Refixing and one- and five-year refixing-risk indicators, adding fixed debt maturing inside the window to floating debt resetting inside the window. The empirical question here is therefore not whether refixing and maturity differ in principle. It is how much the WAM shortcut misses for Portugal at particular horizons, and how that physical refixing comparison differs from shock-weighted fiscal pass-through.

Recent monetary-transmission evidence reinforces the relevance of the timing channel. Johns, Mehrotra and Zampolli [9] study advanced and emerging Europe and show that public-debt maturity affects monetary policy transmission in a horizon-dependent way. Their object is macroeconomic transmission, not a Portuguese debt-manager refixing table, but it makes the same broad point: the maturity structure of public liabilities changes when and how policy-rate shocks matter.

The retail part of the design is closer to the deposit-rate literature than to standard sovereign bond pricing. Drechsler, Savov and Schnabl [6] emphasise that monetary tightening changes the relative return on deposits and therefore the behaviour of deposit holders and banks. Portuguese savings certificates are not bank deposits, but they compete with household deposit rates and are held by households. That makes the subscription margin a plausible place to look for state-dependent pass-through, while also making clear why the paper cannot infer household-level behaviour from aggregate stock data.

The accounting side links the paper back to the companion burden study. Escolano [12] gives the standard debt-dynamics bridge from interest rates and growth to public-debt ratios, while Blanchard [2] shows why that bridge became politically important in a low-rate environment. This paper is about the step before that bridge is used: how much of the inherited stock has actually received the new rate. That step is easy to compress into one maturity statistic, but the compression is exactly where the Portuguese portfolio loses information.

3 Institutional setting

Portugal is an unusually good laboratory for a behavioural repricing test because a large, identifiable share of its debt is held in retail savings instruments — Certificados de Aforro and Certificados do Tesouro — that households may redeem on demand subject to lock-ups and penalties. As of June 2026 these accounted for 15.4 percent of State direct debt. A holder facing a change in the competing return on bank deposits can act on it, so the repricing of that block is in principle a behavioural object rather than a contractual one.

The 2022–2023 tightening provides a clean episode in the everyday sense, not in the econometric sense. Certificados de Aforro grew from EUR 12.9 billion in June 2022 to EUR 32.6 billion by May 2023, a rise of 151 percent. The largest monthly increase in outstanding value was EUR 3,549 million. That is a large movement in a visible instrument class, but it is also one episode, taking place while inflation, household liquidity, deposit competition, and policy communication were all moving.

The direction of the movement is the useful clue. A simple behavioural story would say that tightening speeds pass-through because retail holders redeem low-yielding instruments and reinvest elsewhere at higher rates. Portugal does not fit that story. The certificate's remuneration formula tracked short rates closely enough that the instrument became attractive relative to deposits and drew money in. For the budget, money arriving at the prevailing rate is still repriced money. The channel exists, but it is not the one implied by a redemption hazard.

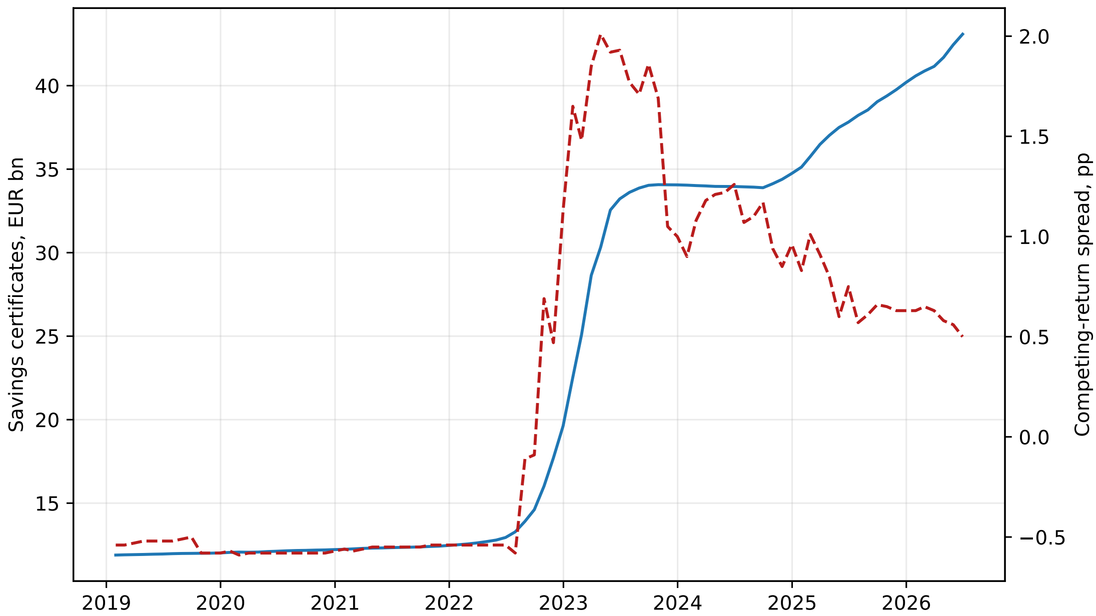

Figure 1: Retail savings certificates and the competing-return spread. The stock is the outstanding amount of Certificados de Aforro; the spread is the policy-rate measure used in the panel less the Portuguese household deposit-rate measure.

Figure 1 is included because the shape of the episode is easier to understand visually than in coefficient form. The stock rises during tightening, then the change in terms offered on new subscriptions arrests the increase. The figure is not causal evidence on its own, but it is enough to discipline the empirical design: the paper should model the subscription margin and should not pretend to have observed gross redemptions.

IGCP's May 2026 investor presentation gives an external check on that reading: its retail net-subscription chart shows the same 2022–2023 inflow episode and subsequent cooling [14]. I use that chart qualitatively. It is not digitised into the regression sample, and it does not replace the missing gross subscription and redemption series that would be needed to estimate a redemption hazard.

4 Data construction

Instrument-class stocks and portfolio indicators come from IGCP historical series, monthly. Competing-return covariates come from the ECB Data Portal. The measurement layer for aggregate debt, GDP, and interest is imported from the companion paper rather than recomputed. The data architecture is intentionally plain: one monthly panel for the repricing question, one annual aggregate dataset for the burden translation, and generated manuscript tables so the paper does not contain hand-typed results. Table 1 summarises the portfolio state used to initialise the kernel. The average maturity input is the IGCP annual-report statistic [13].

The universes are not identical. IGCP composition data describe State direct debt, while the companion burden series are Maastricht general-government debt and national-accounts interest. The fiscal translation is therefore a scale sensitivity, not an independently identified general-government cost. The paper keeps that translation because it is useful for order of magnitude, but it is not used to claim a precise budget effect.

Table 1: Portfolio inputs for the repricing kernel

Quantity	Value
Observation month	June 2026
Average residual maturity (years)	7.52
Fixed-rate share (%)	85.8
Floating or indexed share (%)	14.2
Retail certificate share (%)	15.4
Panel start	February 2001
Panel end	June 2026

Source: author calculations from IGCP monthly stocks and portfolio indicators.

The data are informative, but they do not do more than they say. The unit of observation is a class-month, not a holder. Retail certificate data are published only as aggregate stocks, so the panel cannot separate households that subscribe, redeem, roll over, or remain inactive within the month. The paper therefore avoids individual language. It is a portfolio-flow design, not a household microeconometric design.

The other constraint is that only outstanding value is observed. Gross subscriptions and gross redemptions are not published in the source series. A flat month is consistent with no activity and with large offsetting flows, so a redemption hazard is not identified. A positive month is also not a pure subscription observation. For Savings Certificates the published stock combines subscription principal with capitalised interest, and the capitalisation term itself moves with the remuneration formula. The regression outcome is therefore the positive part of the outstanding-value change, divided by opening stock. It is an observed stock-value measure, not a lower bound on gross repricing, and the coefficient is interpreted only as a descriptive association in that process.

This also means the behavioural block is not a clean opening-stock repricing kernel. Positive retail subscriptions are new financing flows, not euros from the opening stock receiving a new coupon. They are informative about the fiscal rate paid on new retail funding, but they should not be confused with legacy liabilities repricing. The paper therefore treats the retail-flow estimate as a sensitivity rather than as part of the identified central stock kernel.

The contractual side has a different limitation. No dated redemption schedule is published in machine-readable form for the whole stock, so the contractual track uses a stylised retirement profile with the same mean maturity. That is not a reconstruction of IGCP's cash-flow calendar. It is a deliberately transparent counterfactual that lets the paper isolate the composition problem: what changes when part of the stock reprices without maturing?

5 Kernel framework

Let $K_{\text{stock}}(h)$ be the share of the stock outstanding at the horizon origin that carries a repriced rate by horizon h. The notation is simple, but the interpretation is important. A euro can join $K_{\text{stock}}(h)$ because it matures and is reissued, or because it resets while staying outstanding. New retail money arriving on the offered rate is different: it can affect fiscal exposure to the shock, but it is not a repricing event for the opening stock. Putting those objects into one hazard is convenient, but it hides the part of the mechanism this paper is trying to measure.

The contractual component is debt that reprices by maturing and being reissued. This is the component for which a maturity clock is conceptually appropriate. If a redemption schedule is known, it should be used directly. If only a mean maturity is known, a stylised profile is a second-best approximation. The reset component is different. Floating-rate and inflation-linked debt can receive a new rate without leaving the stock, so it is not a survival event. Two assumptions are bundled there and should be separated. The first is *reset timing*: how often the rate is refreshed, taken centrally as one year. The second is *shock loading*: how much of a given shock each instrument absorbs, taken here as one for one. Neither is estimated, and they are not the same object — an inflation-linked coupon does not respond to a policy or market-rate shock the way a Euribor-linked coupon does, and the two need not reset on the same schedule. Treating a reset block as a single one-year, one-for-one block is therefore a scenario choice, and the sensitivity table varies both legs of it separately.

These components have to be defined on *mutually exclusive* shares of the opening stock, which is less obvious than it sounds. Retail and rate type cut across each other in the Portuguese portfolio: Savings Certificates are retail *and* variable-rate, with Series F indexed to three-month Euribor, while Treasury Certificates are retail and carry a guaranteed fixed schedule. An earlier version of this paper subtracted the whole retail block from the fixed-rate track while simultaneously treating the whole non-fixed residual as a reset block, which removed Savings Certificates from a track they were never on and counted them on another. The stock is now partitioned into four classes that sum to one — wholesale fixed, retail fixed, retail variable, wholesale floating — each with its own repricing law, and the construction refuses to run if the shares imply a negative class.

The correction matters less for the total than for the decomposition. At annual horizons the repriced share is unchanged, because the retail block was being given the same contractual profile either way; what changes is that it was being reported in the behavioural column. The behavioural component is now purely a flow: retail money arriving on the prevailing rate, additive to the partition rather than carved out of it. The empirical test below is deliberately modest; it asks only whether the subscription margin carries a detectable response to the competing-return spread.

A benchmark qualification belongs here rather than in a limitations list. Weighted average maturity is the proxy this paper tests, and it is the one the companion study uses, but it is not the only thing Portugal's debt manager publishes. IGCP's risk framework monitors interest-rate refixing explicitly. Its Annual Report carries a refixing profile: the share of the adjusted portfolio due to refix or mature inside each maturity bracket. That profile is a closer published comparator to the object constructed here than a scalar maturity statistic, and the imposed contractual and reset shapes should be checked against it.

That check is now included. IGCP's investor presentation reports the ESDM refixing-risk windows for Portugal at end-March 2026, and Table 2 compares those published windows with both the WAM proxy and the scenario kernel evaluated from the monthly panel state available at that same date. This is a benchmark, not a full calibration. The ESDM chart is cumulative and coarse, and the denominator is the debt manager's adjusted portfolio rather than the exact portfolio basis used in the monthly panel. The model still has to impose reset timing, contractual shape and shock loading. The comparison is useful precisely because it asks whether the scenario kernel is in the neighbourhood of the debt manager's own refixing view before the paper turns to the simpler WAM benchmark.

Table 2: Official ESDM refixing benchmark, WAM proxy, and scenario kernel

Window, years	ESDM share (%)	WAM share (%)	Scenario share (%)	WAM minus ESDM (pp)	Scenario minus ESDM (pp)
0-1	25.2	13.0	19.7	-12.2	-5.5
0-5	50.9	50.0	42.0	-0.9	-8.9

Official shares are Portugal's ESDM refixing-risk windows at 2026-03-31, as published by IGCP. The WAM and scenario columns use the monthly panel state at or before that date and the same cumulative windows. The comparison is qualified by different denominator and portfolio-basis conventions across sources.

The result sharpens rather than removes the paper's caution. The scenario kernel is closer to the official ESDM value in the one-year window, while WAM is closer in the five-year cumulative window. That mixed pattern is the point: a scalar maturity statistic does not identify a repricing path, and even a coarse refixing-risk profile does not identify shock-weighted pass-through without instrument-level reset dates, reference indices, loadings and subscription flows. The published ESDM windows are therefore used as an external check on the imposed shape, not as a substitute for the missing instrument-level schedule.

The benchmark against which we measure is the standard proxy,

$$K^{\text{WAM}}(h) = 1 - (1 - 1/m)^h, \tag{1}$$

with m the published weighted average maturity. Equation (1) is the discrete annual constant-hazard formula used in the companion burden paper. It is memoryless: it implies a long right tail, leaving a substantial share unrepriced after a decade.

The opening-stock scenario kernel is

$$K_{\text{stock}}(h) = K^C(h) + K^R(h), \tag{2}$$

where K^C is contractual repricing and K^R is reset repricing. The fiscal pass-through exposure used in shock translations is

$$P(h, \Delta) = K^C(h) + \lambda K^R(h) + F^N(h, \Delta)/D_0, \tag{3}$$

where λ is the shock loading on reset instruments, $F^N(h, \Delta)$ is new retail funding attracted at the prevailing rate, and D_0 is the opening debt stock. The difference relative to the maturity proxy is then

$$B(h, \Delta) = P(h, \Delta) - K^{\text{WAM}}(h). \tag{4}$$

The decomposition reported in the tables splits B into a shape component and a behavioural component. Neither is an identified estimate. The shape component combines observed portfolio composition with an *imposed* linear retirement track and an *imposed* reset cycle, because the dated redemption schedule is not published; the paper shows below that the sign is robust to the reported reset-timing alternatives but not to weaker shock loading. The behavioural component uses a subscription response that is precisely estimated but not identified. Both are therefore model-based sensitivities. The claim this paper defends is not that the difference equals a particular number, but that weighted average maturity and a mean do not determine a unique repricing path: plausible composition, timing, loading and new-funding assumptions generate materially different paths, of different signs relative to the constant-hazard benchmark.

6 Estimation

The specification was frozen before fitting and estimated once; no specification search was performed. The outcome is the repriced share of the opening stock. The regressors are the signed parts of the lagged competing-return spread, an indicator for the June 2023 change in the terms offered on new subscriptions, and portfolio composition. Standard errors are Newey–West. Estimation is on the two retail classes aggregated to a single monthly series, 304 months. That aggregation is not cosmetic. Stacking the classes places every month of one class before every month of the other, so calendar time jumps backwards at the join: a twelve-month bootstrap block can straddle a fifteen-year gap, and the Newey–West lag structure does not correspond to month-to-month dependence. Aggregating also weights each class by the euros it carries rather than giving a small class and a large one equal influence, which removes the need for class effects.

The spread-widening coefficient is +0.0214 with a standard error of 0.0061 ($p = 0.00$). Taken alone that is a detectable positive association between the competing-return spread and retail repricing. Two things stop it supporting a causal reading.

First, the asymmetry the design turns on — whether widening and narrowing load differently — has a bootstrap interval of [-0.016, +0.046] across 1,000 moving-block replicates and still covers zero. The directional claim the study was built to test remains unresolved.

Second, and more seriously, the falsification test no longer passes cleanly. The fixed-rate share is a portfolio statistic no household observes and should carry no weight; it loads at -0.00051 ($p = 0.07$). That is close enough to conventional significance to be a warning rather than a reassurance. The natural reading is that the monthly series carries common time variation which the specification cannot separate from the spread. The coefficient above should therefore be read as descriptive — an association in the observable positive outstanding-value change process — and not as an identified behavioural response. Nothing downstream in this paper treats it as identified: the kernel's central behavioural effect is set to zero and the fitted response enters only as a sensitivity band.

That fragility is not surprising. The sample contains one large widening episode, so its timing cannot be separated cleanly from everything else moving with it. The certificate's contractual remuneration rate is not published machine-readably, so a short-rate index stands in and attenuation of unknown size is expected. The outcome also mixes accounting and behaviour. Positive outstanding-value change can reflect new principal, reissued money, or capitalised interest, while net-outflow months tell us that money left but not how much gross subscription and redemption occurred underneath. A more flexible specification would make the table look more elaborate without creating the missing gross-flow data or separating those accounting components.

Table 3: Subscription-margin regression

Term	Coefficient	Newey–West s.e.	p-value
const	-0.0130	0.0056	0.02
spread_widening_pp	0.0214	0.0061	0.00
spread_narrowing_pp	-0.0007	0.0016	0.66
post_policy_break	-0.0250	0.0079	0.00
average_residual_term_years	0.0031	0.0010	0.00

Outcome: positive outstanding-value change as a share of opening class stock. Source: author calculations from IGCP and ECB data.

Table 3 is deliberately shown in full because a compressed "behavioural response" number would overstate what the data can support. The policy break and maturity covariate are statistically precise in this frozen descriptive specification, but precision is not identification. The observed stock-value outcome, the short-rate proxy and the placebo loading mean those coefficients should not carry a structural household-behaviour interpretation. The behavioural component can be carried through the kernel as a sensitivity, not as the centrepiece of the paper.

7 The kernel and the WAM comparison

Table 4 reports the scenario-minus-WAM difference at a 100 basis-point shock with the central behavioural effect set to zero. The sign is positive at one year and negative at the longer reported horizons, so the table should be read as a bounded model exercise rather than as an identified empirical correction in one direction. The difference combines an early reset effect, a stylised contractual-shape effect, and behavioural bounds reported separately.

Table 4: Scenario-minus-WAM repricing differences, +100 bps

Horizon	Proxy share (%)	Scenario exposure (%)	Difference (pp)	Shape (pp)	Behaviour (pp)
1	13.30	19.89	6.60	6.60	0.00
3	34.83	31.31	-3.52	-3.52	0.00
5	51.01	42.72	-8.29	-8.29	0.00
10	76.00	71.25	-4.75	-4.75	0.00

Source: author calculations. Shape and behaviour sum to the reported pass-through exposure difference.

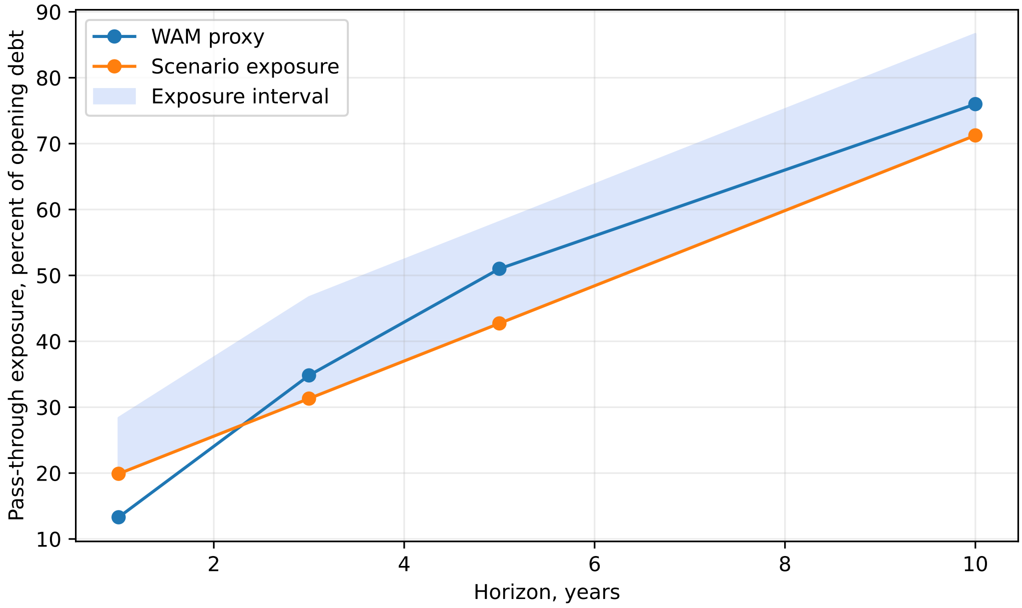

Figure 2: Weighted-average-maturity proxy and composition-sensitive pass-through exposure. The interval is the generated behavioural interval around that exposure. The scenario is model-implied under explicit portfolio assumptions, not estimated: its contractual profile, reset cycle and shock loading are imposed.

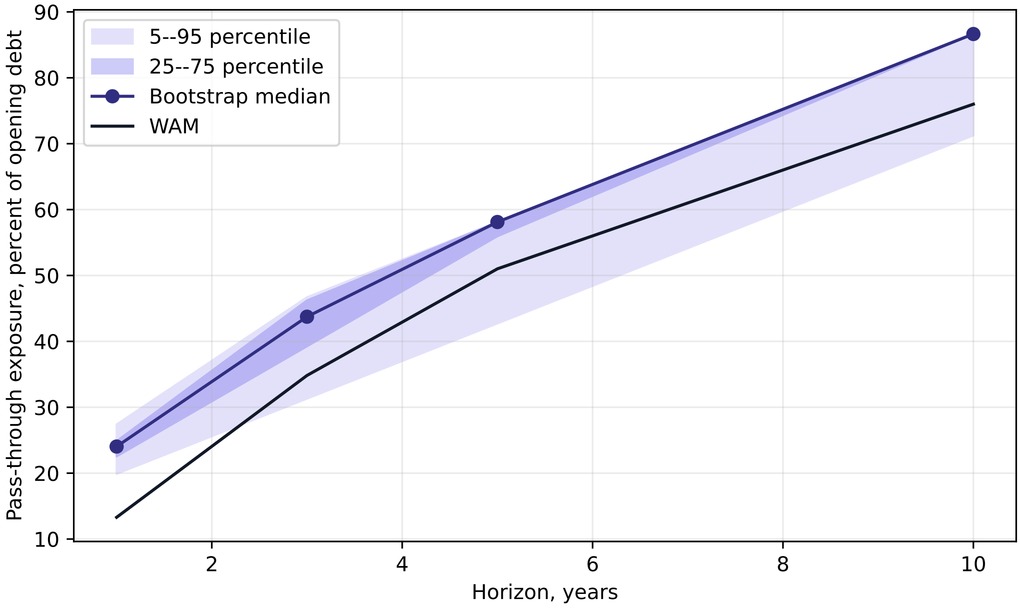

Figure 3: Bootstrap uncertainty around scenario pass-through exposure. The contractual profile, reset timing and shock loading are imposed; the bands propagate only the moving-block bootstrap draws for the behavioural response.

The mechanism is compositional, but not in a single piece. At one year the shape component contributes 6.60 percentage points and requires no behavioural assumption. That number combines the positive reset effect from the non-fixed share with the negative difference between a stylised linear retirement profile and the constant-hazard benchmark. Equation (1) cannot represent reset without exit, but the shape term should not be narrated as a pure reset effect.

Two qualifications. First, the shape term is *not monotone* in the horizon: it turns negative by three and five years, because beyond the reset window the memoryless tail retires more than a profile that has already retired its fast component. "The constant-hazard kernel is too slow" is true early and false later under the zero-behaviour central case. Second, the behavioural component's band runs from zero at every horizon, because the estimate behind it is not identified, for the reasons in Section 6. We report behavioural upside only as a sensitivity; a point estimate without that caveat would be precisely the failure this paper criticises.

The fiscal translation is reported only as a scale sensitivity because the composition kernel is built on State direct debt while the burden denominator is general-government GDP and debt.

Figure 2 makes the non-monotonicity visible. The central kernel is above the WAM proxy at one year and below it at middle horizons. The reset block brings pass-through forward; the stylised contractual profile can then sit below the memoryless tail once the fast component has already repriced. That pattern is the reason for using the phrase "compositionally sensitive" rather than the easier phrase "too slow". The standard proxy is not wrong in one direction at every point of the curve; it is using the wrong clock for part of the stock.

Figure 3 adds the empirical uncertainty that the point curve alone hides. The band is not mainly a story about debt managers changing contractual maturities; those inputs are held fixed. It is the uncertainty from the retail-flow estimate carried through the kernel. That is why the band is useful even though the behavioural coefficient is not identified. It shows which part of the measurement correction is stable composition and which part is still a thin-data behavioural estimate.

8 Pass-through

Simulating forward under the scenario kernel also clarifies the denominator problem. The companion paper holds nominal GDP fixed across the ten-year horizon to isolate refinancing arithmetic. That convention is transparent, but it is a scenario, not a forecast. At a 100 basis-point shock and a five-year horizon, the incremental burden is 0.383 percent of GDP under the frozen-GDP assumption and 0.315 percent under an illustrative central nominal-growth path. The difference is a denominator sensitivity, not an empirical correction.

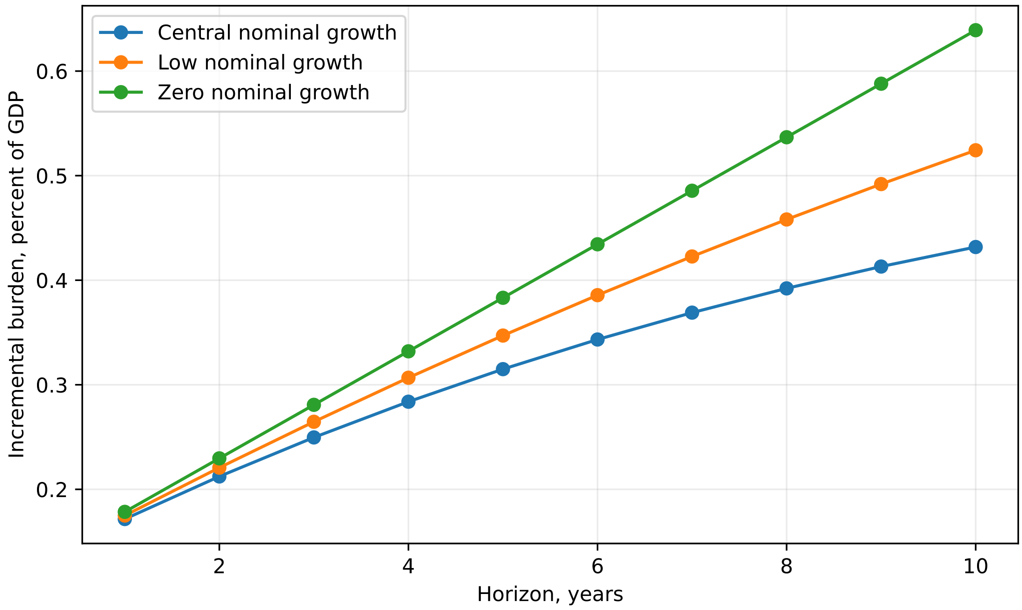

Figure 4: Incremental burden under a +100 basis-point shock across nominal-GDP paths. The lines differ only in the denominator path.

Table 5: Half-life of incremental pass-through under central nominal growth

Shock (bps)	Maximum burden effect (% GDP)	Half-life (years)
-200	0.864	3.0
-100	0.432	3.0
-50	0.216	3.0
50	0.216	3.0
100	0.432	3.0
200	0.864	3.0

The half-life is the first horizon at which the absolute burden effect reaches half of its simulated maximum.

We note one artefact so it is not mistaken for a finding. Simulated half-lives differ by shock direction, but that asymmetry is manufactured by clipping the behavioural response at zero, which prevents a rate fall producing a negative contribution. It is a modelling choice, and the estimated asymmetry does not separate from zero.

The pass-through exercise therefore has two separate lessons. The kernel decides how much of the stock has received the shock. The GDP path decides how large that same euro interest change looks as a burden ratio. A fiscal note that freezes GDP is useful for comparability with full-pass-through arithmetic, but it should not be read as a central projection. The economic object is a euro interest bill; the political and fiscal object is usually the bill as a share of GDP. The two should not be allowed to drift into each other.

9 Uncertainty and sensitivity

The deterministic paths above are useful because they isolate one mechanism at a time. They are not enough by themselves. A reader also needs to know whether the result survives plausible movements in the modelling choices. The paper therefore adds two generated checks: a Monte Carlo fan chart for the burden path and a one-at-a-time sensitivity table for the kernel assumptions.

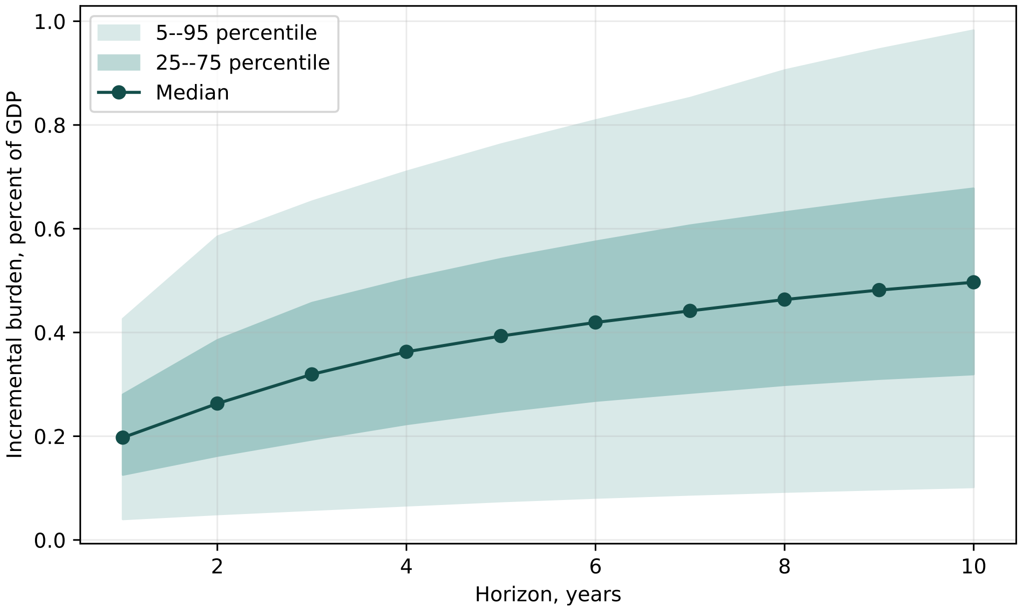

Figure 5: Illustrative stochastic sensitivity for incremental burden pass-through. The shock and growth distributions are chosen, not estimated, so the band is a percentile range of those assumed inputs and not a forecast or confidence interval.

Figure 5 uses 2,000 generated draws. The mean shock in the draw set is 99 basis points and the mean nominal-growth draw is 4.0 percent. By the fifth year, the simulated burden effect has a median of 0.393 percent of GDP, with a 5–95 percentile interval from 0.075 to 0.763. The wide band is not a failure of the kernel; it is the fiscal reality that the same repricing path can look quite different once the shock size and GDP denominator are allowed to vary.

The band must not be read as a forecast interval. The shock and growth distributions are author-chosen rather than estimated or externally calibrated, so the 5–95 range describes the spread of those assumed inputs, not empirical uncertainty. It is an illustrative stochastic sensitivity; the deterministic envelopes above carry the substantive comparisons.

Table 6: Sensitivity of the kernel to modelling assumptions, +100 bps

Scenario	One-year difference (pp)	Five-year pass-through exposure (%)	Five-year burden (% GDP)
Central kernel	6.60	42.72	0.383
Slow reset (two-year cycle)	6.18	42.72	0.383
Fast reset (six-month cycle)	6.60	42.72	0.383
Reset loading 0.5	-0.50	35.62	0.320
Reset loading 0.25	-4.05	32.08	0.288
Memoryless contractual shape	12.30	57.96	0.520
Behaviour off	6.60	42.72	0.383
Behaviour low case	6.60	42.72	0.383
Behaviour high case	15.09	58.11	0.521

The table varies one assumption at a time around the central kernel. Source: author simulations.

Table 6 asks a different question. Instead of randomising the macro environment, it moves the main modelling assumptions one at a time. The one-year difference range across those checks is 19.13 percentage points. Two rows matter most because they attack the compositional claim directly, and they do so through different assumptions. If reset instruments are assumed to take longer to receive the new rate, the first-year correction remains positive in the reported slow-reset and fast-reset alternatives, although its size moves. The sign changes only when the reset block receives the new rate on schedule but absorbs less of the shock: the reset-loading rows hold timing fixed and vary transmission alone, and that is enough to move the one-year result from positive to negative. Timing and loading are therefore both first-order, which is the sharpest form of the paper's argument — a published maturity statistic constrains neither, so it cannot pin down the sign of the one-year correction, let alone its size. The behavioural rows matter less for the central conclusion. They widen or narrow the total, but they do not remove the reason the WAM proxy is incomplete. The defensible claim is therefore about non-identification by WAM alone, not systematic understatement at every short horizon.

10 Implications for fiscal reporting

The easiest way to use this result is not to replace weighted average maturity with a new black box. The useful change is smaller and more practical: report a short repricing bridge beside the usual maturity statistic. The bridge should state how much of the opening stock is fixed-rate and maturity-driven, how much is reset-linked, and how much belongs to retail or other instruments whose stock can move with offered terms. A reader can then see why the same published maturity can imply different first-year budget exposure across countries or across years.

This also changes how uncertainty should be presented. A conventional sensitivity table often gives a clean interest-cost number for a rate shock and then leaves the timing assumption implicit. In the Portuguese case the more honest presentation would separate the contractual component, which is mostly a portfolio-accounting issue, from the behavioural component, which is genuinely uncertain in published data. That distinction matters because it tells the reader which part of the number is anchored in the stock and which part is a scenario.

The retail result is particularly useful for policy communication. It would be tempting to say that households accelerate pass-through by fleeing old retail debt when market rates rise. The Portuguese episode points the other way: households subscribed when the certificate return became attractive. The fiscal problem was not that existing holders had to be replaced at a higher rate; it was that new retail funding arrived on terms tied to the new rate environment. That is still pass-through, but the language is different and the policy lever is different.

The final implication concerns publication practice. If gross subscriptions and redemptions were published in a stable machine-readable series, the behavioural block could be estimated directly rather than treated as a bound. The paper therefore has a data-policy conclusion as well as a modelling one: routine publication of gross retail flows would materially improve public understanding of interest-cost pass-through.

11 Conditional historical validation

This exercise is conditional historical validation, not a forecast. Realised future rate paths are fed into it on purpose, so that kernel timing error can be separated from yield-path error, and a forecast would not have them. What it can answer is narrower and still worth asking: given what rates actually did, did this kernel time the pass-through better than the maturity proxy?

We freeze the specification, reconstruct the portfolio state as it stood at each cut date, and roll forward from that state. The rate fed forward is the realised *ten-year benchmark* yield, so that kernel error is isolated from yield-path error. It is not an issuance-weighted funding rate: the securities repricing in a given year are not all ten-year, so the exercise measures the timing kernel against a common reference path rather than reproducing actual issuance cost. Repricing is accumulated by cohort, so debt that repriced in an earlier year carries the yield it repriced at rather than being revalued at the latest one. The realised effective rate is imported from the companion paper.

The behavioural track is driven by the realised competing-return spread, which is the variable the coefficient was estimated against. That correspondence is not cosmetic. An earlier version drove it with the gap between the ten-year benchmark and the cut-date effective rate, which is a different object in different units: a coefficient expressed in repriced share per point of competing-return spread was being applied to something closer to a term premium. Where the spread is unavailable the behavioural track is set to zero rather than driven by a stand-in.

Table 7: Conditional historical validation errors

Cut year	Model	Mean abs. error (bps)	Worst error (bps)	N
2014	Scenario kernel	40.35	79.31	11
2014	WAM benchmark	43.24	82.09	11
2014	Immediate full pass-through	95.09	178.70	11
2014	Random walk	120.75	197.39	11
2018	Scenario kernel	12.50	30.01	7
2018	WAM benchmark	12.92	27.08	7
2018	Random walk	63.62	104.35	7
2018	Immediate full pass-through	119.67	178.70	7
2021	WAM benchmark	9.81	23.17	4
2021	Scenario kernel	10.08	23.62	4
2021	Random walk	24.52	32.45	4
2021	Immediate full pass-through	81.05	115.72	4

Errors are basis points on the realised average debt rate. Source: author calculations.

Table 8: Model comparison across validation cut years

Model	Mean abs. error (bps)	Worst error (bps)	Wins	Cuts
Scenario kernel	20.98	79.31	2	3
WAM benchmark	21.99	82.09	1	3
Random walk	69.63	197.39	0	3
Immediate full pass-through	98.60	178.70	0	3

Wins count cut years in which the model has the lowest mean absolute error. Source: author calculations.

The scenario kernel has the lower mean absolute error at two of the three cut dates, by small and unstable margins: 2.88 basis points at the 2014 cut and 0.42 basis points at the 2018 cut, while the WAM benchmark is better at 2021 by 0.27 basis points. That ordering should not be read as forecasting skill. Three cut dates with between four and eleven annual observations each, separated by margins this size, cannot distinguish two timing assumptions.

The ordering has also moved every time an implementation defect was corrected, which is itself the most useful thing this section reports. Earlier versions had the kernel winning two of three, then losing all three, then winning one, before returning to two of three — successively because the kernel was built at zero shock, which silences the behavioural channel whatever response is supplied; because each year's yield was applied to the whole cumulative repriced share rather than to the cohort that repriced; because the portfolio state came from the end of the sample rather than the cut; and because the behavioural coefficient was driven by the wrong variable. A ranking that is this sensitive to implementation is not a ranking a reader should rely on, and we would rather say so than present the current ordering as a result.

This is consistent with the rest of the paper rather than in tension with it. The compositional difference is concentrated in the first year, and a backtest over the available annual observations has little power to detect it against effective-rate volatility. The backtest is still necessary because it prevents the kernel from being sold as a superior forecasting model when its strongest evidence is a measurement decomposition.

12 Discussion and limitations

The contribution is a measurement result: the standard pass-through proxy does not identify a unique timing path once composition and reset assumptions are opened up. Anyone using weighted average maturity to time a rate shock through a budget should report the share of the stock that may reprice without exiting, and should show the assumptions behind that share. This is not a refinement for the sake of refinement. A first-year budget exercise is exactly where floating and indexed liabilities matter, and that is exactly where a maturity-only proxy is least able to see them.

The behavioural thesis is not supported in the strong form. The channel is visible in the raw data — a 151 percent stock increase and an abrupt stop at a dated policy change — but it is not identifiable at the precision published sources allow, principally because gross retail flows are not published and the contractual remuneration rate is not machine-readable. That is a useful negative result. It prevents the paper from turning an attractive Portuguese institutional episode into a broad claim about household portfolio elasticities.

The limitations are therefore part of the argument rather than a disclaimer at the end. The contractual track rests on a stylised retirement profile in the absence of a published schedule; the unit of observation is a class-month, not a holder; and the sample contains a single large tightening episode. The model also treats reset instruments as repricing within the first year rather than modelling their exact reset dates. The four-way split between wholesale fixed, wholesale floating, retail fixed and retail variable instruments is now explicit and guarded in the code, so the remaining limitation is not an unresolved overlap in the opening-stock categories. It is the coarseness of the published aggregates: a stronger design would need ISIN- or cohort-level reset schedules, gross retail subscriptions and redemptions, and observed repricing weights.

13 Conclusion

Weighted average maturity does not uniquely determine the speed at which a rate shock reaches a sovereign interest bill. For Portugal, plausible one-year repricing shares are materially affected by assumptions about reset timing, maturity shape, and retail flows. A behavioural channel may exist and appears opposite to the intuitive redemption story, but the published aggregate data do not identify it.

The practical recommendation is simple. Start from weighted average maturity only for the part of the stock whose repricing is tied to maturity. Put floating-rate and indexed debt on a reset track, put retail instruments on a separate subscription and redemption track where gross flows exist, and then translate euro interest effects into burden ratios under explicit GDP paths. That workflow is more demanding than the one-number WAM proxy, but it prevents the proxy from assigning the wrong timing to instruments whose repricing clock is not a maturity clock.

Data availability

All inputs are public. Instrument-class stocks and portfolio indicators come from IGCP historical series; competing-return covariates from the ECB Data Portal; aggregate debt and interest from Eurostat. The aggregate series are those of the companion interest-burden study [1], used unchanged.

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